

# SHUVAM PAL

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## Education

Dayananda Sagar College of Engineering

Bengaluru, India

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI** 

Dec 2025 – Feb 2026

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions*

*Remote*

- Built a resilient **Puppeteer/Node.js workflow engine** for MCA company-registration filings, reducing manual filing time by **85%** through state recovery, retries, and public-site API lookups.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker** and automated **GitHub Actions** deployments to **Google Cloud Run**, improving release velocity by **70%**.

**Google Summer of Code** 

May 2025 – Sept 2025

*Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Built MRI segmentation tooling for **95 anatomical regions**, turning model outputs into usable masks and 3D visualization artifacts inside an existing application workflow.
- Reduced processing runtime by **87%** with async/parallel execution and fewer repeated post-processing passes over large MRI volumes.
- Integrated outputs into 3D reconstructed surface visualization flows so model results stayed inspectable and useful to end users.

## Open Source

[Contributions](#) 

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working with existing maintainers, code style, and review constraints.

## Projects

**Swish Support System** | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built an AI support system with per-conversation context for active item, issue type, requested resolution, pending confirmations, and evidence artifacts, achieving **85%+** policy-correct resolutions on a **50-case** eval set.
- Reduced compensation leakage by **25%** versus a naive refund baseline by separating semantic understanding from deterministic policy and tracing high-risk cases with **Langfuse**.

**Murdoc** | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built a self-hosted **AI security gateway** for LLM, HTTP tool, and MCP traffic, blocking **90%+** of prompt-injection and unsafe-tool attempts across a **150-case attack corpus** with false positives under **5%**.
- Added route profiles, RBAC runtime settings, PII redaction, allow/block rules, **OpenTelemetry** metrics/traces, health checks, and audit ledgers for **100%** of policy decisions.

**Penny Lane Capital (Tradeage)** | *Python, Agno, Pydantic, SQLite, yfinance, JSONL*

- Built an **Agno Workflow** multi-agent research system with analyst roles, bull/bear debate, trader, risk review, and portfolio approval using Pydantic state, checkpoints, memory, and JSONL traces.
- Evaluated agent decisions with historical-market backtests, reward loops, validation harnesses, and **5 baselines**, producing return, drawdown, and win-rate scorecards across market windows.

## Technical Skills

**Languages:** Python, Go, TypeScript

**Frameworks:** FastAPI, Gin, Echo, Node.js, Agno

**Tools:** PostgreSQL, Redis, RabbitMQ, Docker, Langfuse, OpenTelemetry, GitHub Actions, AWS